

Avaneesh Sanjay Bhoite

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PROFESSIONAL SUMMARY

Computer Science graduate student at Emory University with hands-on experience building AI systems, cloud-native architectures, and machine learning pipelines. Developed a serverless multi-agent platform on AWS at TCS, built ML models for financial forecasting and medical image classification, and published research on reinforcement learning. Comfortable working across the full stack from model development to cloud deployment.

EDUCATION

Emory University

Atlanta, GA

Master's in Computer Science. GPA: 3.8/4.00.

Aug 2025 - Present

Relevant Coursework: Machine Learning and Computational Analytics, Information Retrieval and AI Systems, PhD-level Algorithms, Data Mining, Database Systems, NLP, Human-AI Interaction, Human-Computer Interaction (HCI), Software Design

Dr. Vishwanath Karad MIT World Peace University

Pune, India

B.Tech in Computer Science and Engineering. GPA: 3.8/4.00.

Jul 2024

SKILLS

Machine Learning & AI:	Generative AI, LLMs, Vector Databases, Prompt Engineering
Libraries and Frameworks:	TensorFlow, scikit-learn, Pandas, NumPy, PyTerrier, Seaborn, Django, Flask
Cloud and Tools:	AWS (Lambda, DynamoDB, S3, SQS), Amazon Bedrock, VS Code, Bootstrap, Android Studio, Linux, MacOS, Windows
Programming and Databases:	Python, JavaScript, Java, HTML/CSS, Node.js, React.js, MySQL, MongoDB

EXPERIENCE

Agentic AI Intern

May 2026 - Aug 2026

Tata Consultancy Services (TCS)

Edison, NJ

- Manual peer review of SQL report-logic changes took 24-48 hours per ticket across a team of 50-100, the top bottleneck in Prudential's deployment pipeline; designed and built SuPRvisor, a serverless AI platform (AWS Lambda, DynamoDB, S3, SQS, Amazon Bedrock) to automate it.
- Built a multi-agent Lambda architecture and custom intake workflow to fully automate semantic SQL review, blank-data risk assessment, and executive review generation, replacing what had been a manual process.
- Implemented automated data-quality checks, report profiling, and metadata validation, generating structured Excel audit packages with remediation guidance in place of manual audits.
- Engineered DynamoDB-based persistence for workflow tracking, Bedrock response caching, and developer performance analytics, cutting per-ticket review time from 24-48 hours to 5-10 minutes -- over a 99% reduction -- while automating 100% of the review process.

Software Development Engineer

Jan 2024 - Jul 2024

FairShare IT Services

Pune, India

- Engineered Python predictive models (Pandas, TensorFlow, scikit-learn) to forecast prices for 100+ stocks from historical data, achieving 96% R2 and 90% directional accuracy in backtesting.
- Automated data collection and candidate-stock screening pipelines via targeted web scraping (ElementTree) and report generation (Docx), removing manual research work in building the model's input universe.

Research Co-Developer

Jul 2022 - Mar 2023

Skoda Auto Volkswagen Pvt. Ltd.

Pune, India

- Dealership service teams lacked consistent onboarding and performance tracking; collaborated on an 8-person team to

build interactive training software and KPI-driven instructional modules adopted by 6 Service Managers, extending training to teams of 40-50 employees each (240+ employees reached).

PROJECTS

Oral Cavity Cancer Classification Model

Sep 2025 - Present

Research Intern

- Developed an end-to-end Python pipeline using h5py to extract features from large-scale Whole Slide Images (WSI) and train a Logistic Regression classifier to distinguish tumor patches; processed up to 20 full-sized .svs files (6-8 GB per slide) yielding patch counts in the 10,000 range.
- Engineered data processing scripts mapping patch coordinates to pathological binary masks, automatically generating training labels via pixel-level tumor density thresholds.
- Built interpretability modules using PIL and NumPy to overlay color-coded prediction grids onto slide thumbnails, enabling visual verification of tumor classifications.

Unity ML-Agents: Gaming through Reinforcement Learning

Jan 2024 - Jun 2024

Team Leader

- Developed a computer racing game integrating reinforcement learning via the Unity ML library to dynamically adapt non-playable characters (NPCs) to player behaviors.
- Simulated personalized challenges with an 87.5% success rate; presented and published the associated research paper at the IEEE 2nd WCONF 2024.